



JKBOSE 2026

CLASS 11TH

LINEAR INEQUALITIES

MATHS

LECTURE- 03

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TOPICS TO BE COVERED

1

Exercise Discussion 5.1

EXERCISE – 5.1

QUESTION

Solve $24x < 100$, when

(i) x is a natural number

(ii) x is an integer.

(Sol) Given that $24x < 100$

$$\Rightarrow x < \frac{100}{24}$$

(i) Since x is a natural number.

$$\text{So, solution set} = \{1, 2, 3, 4\}$$

(ii) Since x is an integer.

$$\text{So, solution set} = \{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4\}$$

QUESTION

Solve $-12x > 30$, when

(i) x is a natural number

(ii) x is an integer. *h/w.*

QUESTION

Solve $5x - 3 < 7$, when

- (i) x is an integer.
- (ii) x is a real number.

H/w.

QUESTION

Solve $3x + 8 > 2$, when

(i) x is an integer.

(ii) x is a real number.

(Sol) Given that $3x + 8 > 2$
 $\Rightarrow 3x > 2 - 8$
 $\Rightarrow 3x > -6$
 $\Rightarrow x > -\frac{6}{3} = -2$
 $\Rightarrow x > -2$

(i) Since x is an integer. So,
the solution set = $\{-1, 0, 1, 2, 3, 4, \dots\}$

(ii) Since x is a real number. So,
the solution set = $(-2, \infty)$.

QUESTION

Solve the inequalities in Exercises 5 to 16 for real x .

$$4x + 3 < 5x + 7$$

#.

QUESTION



Solve the inequalities in Exercises 5 to 16 for real x .

$$3x - 7 > 5x - 1$$

HP/W

QUESTION



Solve the inequalities in Exercises 5 to 16 for real x .

$$3(x - 1) \leq 2(x - 3)$$

(Sol) Given that $3(x - 1) \leq 2(x - 3)$

$$\Rightarrow 3x - 3 \leq 2x - 6$$

$$\Rightarrow 3x - 2x \leq -6 + 3.$$

$$\Rightarrow x \leq -3$$

Therefore the Solution Set = $(-\infty, -3]$

QUESTION



Solve the inequalities in Exercises 5 to 16 for real x .

$$3(2 - x) \geq 2(1 - x)$$

HP/W

QUESTION



Solve the inequalities in Exercises 5 to 16 for real x .

$$x + \frac{x}{2} + \frac{x}{3} < 11$$

H/w.

QUESTION



Solve the inequalities in Exercises 5 to 16 for real x .

$$\frac{x}{3} > \frac{x}{2} + 1$$

H/W

QUESTION



Solve the inequalities in Exercises 5 to 16 for real x .

$$\frac{3(x-2)}{5} \leq \frac{5(2-x)}{3}$$

(Sol) Given that $\frac{3(x-2)}{5} \leq \frac{5(2-x)}{3}$

$$\Rightarrow 9(x-2) \leq 25(2-x)$$

$$\Rightarrow 9x - 18 \leq 50 - 25x$$

$$\Rightarrow 9x + 25x \leq 50 + 18$$

$$\Rightarrow 34x \leq 68$$

$$\Rightarrow x \leq \frac{68}{34}$$

$$\Rightarrow x \leq 2$$

Therefore the solution set = $(-\infty, 2]$

QUESTION



Solve the inequalities in Exercises 5 to 16 for real x .

$$\frac{1}{2} \left(\frac{3x}{5} + 4 \right) \geq \frac{1}{3} (x - 6)$$

HP/W

QUESTION



Solve the inequalities in Exercises 5 to 16 for real x .

$$2(2x + 3) - 10 < 6(x - 2)$$

hp/w

QUESTION



Solve the inequalities in Exercises 5 to 16 for real x .

$$37 - (3x + 5) \geq 9x - 8(x - 3)$$

(Sol) Given that $37 - (3x + 5) \geq 9x - 8(x - 3)$

$$\Rightarrow 37 - 3x - 5 \geq 9x - 8x + 24$$

$$\Rightarrow 32 - 3x \geq x + 24$$

$$\Rightarrow -3x - x \geq 24 - 32$$

$$\Rightarrow -4x \geq -8$$

$$\Rightarrow x \leq \frac{-8}{-4} = 2$$

$$\Rightarrow x \leq 2$$

Therefore the Solution Set = $(-\infty, 2]$.

QUESTION



Solve the inequalities in Exercises 5 to 16 for real x .

$$\frac{x}{4} < \frac{(5x - 2)}{3} - \frac{(7x - 3)}{5}$$

HP/W

QUESTION



Solve the inequalities in Exercises 5 to 16 for real x .

$$\frac{(2x-1)}{3} \geq \frac{(3x-2)}{4} - \frac{(2-x)}{5}$$

(Sol) Given that:

$$\begin{aligned} & \left(\frac{2x-1}{3} \right) \geq \left(\frac{3x-2}{4} \right) - \left(\frac{2-x}{5} \right) \\ \Rightarrow & \left(\frac{2x-1}{3} \right) \geq \frac{5(3x-2) - 4(2-x)}{20} \\ \Rightarrow & \left(\frac{2x-1}{3} \right) \geq \left(\frac{15x-10-8+4x}{20} \right) \end{aligned}$$

$$\Rightarrow \left(\frac{2x-1}{3} \right) \geq \left(\frac{19x-18}{20} \right)$$

$$\Rightarrow 20(2x-1) \geq 3(19x-18)$$

$$\Rightarrow 40x - 20 \geq 57x - 54$$

$$\Rightarrow 40x - 57x \geq -54 + 20$$

$$\Rightarrow -17x \geq -34$$

$$\Rightarrow x \leq \frac{-34}{-17}$$

$$\Rightarrow x \leq 2$$

Therefore the Solution Set = $(-\infty, 2]$

QUESTION

Solve the inequalities in Exercises 17 to 20 and show the graph of the solution in each case on number line

$$3x - 2 < 2x + 1 \quad \text{R/W}$$

QUESTION



Solve the inequalities in Exercises 17 to 20 and show the graph of the solution in each case on number line

$$5x - 3 \geq 3x - 5$$

h/w

QUESTION



Solve the inequalities in Exercises 17 to 20 and show the graph of the solution in each case on number line

$$3(1 - x) < 2(x + 4)$$

h/w

QUESTION



Solve the inequalities in Exercises 17 to 20 and show the graph of the solution in each case on number line

$$\frac{x}{2} \geq \frac{(5x - 2)}{3} - \frac{(7x - 3)}{5}$$

R/w

QUESTION

Ravi obtained 70 and 75 marks in first two unit test. Find the minimum marks he should get in the third test to have an average of at least 60 marks.

(Sol) let the marks scored by Ravi in 3rd test = x .

Then according to the question, we get;

$$\left(\frac{70 + 75 + x}{3} \right) \geq 60$$

$$\Rightarrow \left(\frac{145 + x}{3} \right) \geq 60$$

$$\Rightarrow 145 + x \geq 60 \times 3$$

$$\Rightarrow 145 + x \geq 180$$

$$\Rightarrow x \geq 180 - 145.$$

$$\Rightarrow x \geq 35$$

Therefore the req. minimum marks = 35.

Hence the result.

QUESTION

To receive Grade 'A' in a course, one must obtain an average of 90 marks or more in five examinations (each of 100 marks). If Sunita's marks in first four examinations are 87, 92, 94 and 95, find minimum marks that Sunita must obtain in fifth examination to get grade 'A' in the course.

(Sol) let the marks scored by Sunita in the 5th examination = x .

Then according to the Question;

$$\left(\frac{87+92+94+95+x}{5} \right) \geq 90$$

$$\Rightarrow \frac{368+x}{5} \geq 90$$

$$\Rightarrow 368+x \geq 450$$

$$\Rightarrow x \geq 450-368$$

$$\Rightarrow x \geq 82$$

Therefore the required minimum marks = 82.

Hence the result.

QUESTION



Find all pairs of consecutive odd positive integers both of which are smaller than 10 such that their sum is more than 11.

(Sol) let the required pair of consecutive odd positive integers be $(x, x+2)$

Then, According to the Question,

$$x+2 < 10$$

$$\Rightarrow x < 8 \text{ --- (1)}$$

$$\& \quad x + (x+2) > 11$$

$$\Rightarrow 2x+2 > 11$$

$$\Rightarrow 2x > 9$$

$$\Rightarrow x > \frac{9}{2}$$

$$\Rightarrow x > 4.5 \text{ --- (2)}$$

From (1) & (2), we get,

$$4.5 < x < 8$$

Therefore the required pairs are $(5, 7)$ & $(7, 9)$

Hence the result.

QUESTION

Find all pairs of consecutive even positive integers, both of which are larger than 5 such that their sum is less than 23.

(Sol) let the required pair of consecutive even positive integers be $(x, x+2)$

Then, according to the Question, we get

$$x > 5 \quad \text{--- (1)}$$

$$x + (x+2) < 23$$

$$\Rightarrow 2x + 2 < 23$$

$$\Rightarrow 2x < 21$$

$$\Rightarrow x < 10.5 \quad \text{--- (2)}$$

From (1) & (2), we get;

$$5 < x < 10.5 \quad (x = 6, 8, 10)$$

Therefore the required pairs are $(6, 8)$, $(8, 10)$ & $(10, 12)$

Hence the result.

QUESTION

The longest side of a triangle is 3 times the shortest side and the third side is 2 cm shorter than the longest side. If the perimeter of the triangle is at least 61 cm, find the minimum length of the shortest side.

(Sol) let the length of the shortest side = x cm.

$\Rightarrow$ length of longest side = $3x$ cm.

& the length of the third side = $(3x - 2)$ cm

Then, according to the Question, we get;

$$x + 3x - 2 + 3x \geq 61$$

$$\Rightarrow 7x - 2 \geq 61$$

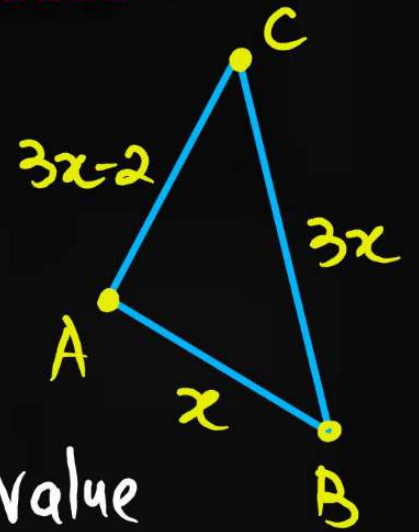
$$\Rightarrow 7x \geq 63$$

$$\Rightarrow x \geq \frac{63}{7} = 9$$

$$\Rightarrow x \geq 9$$

Therefore the required minimum value
= 9 cm.

Hence the result.



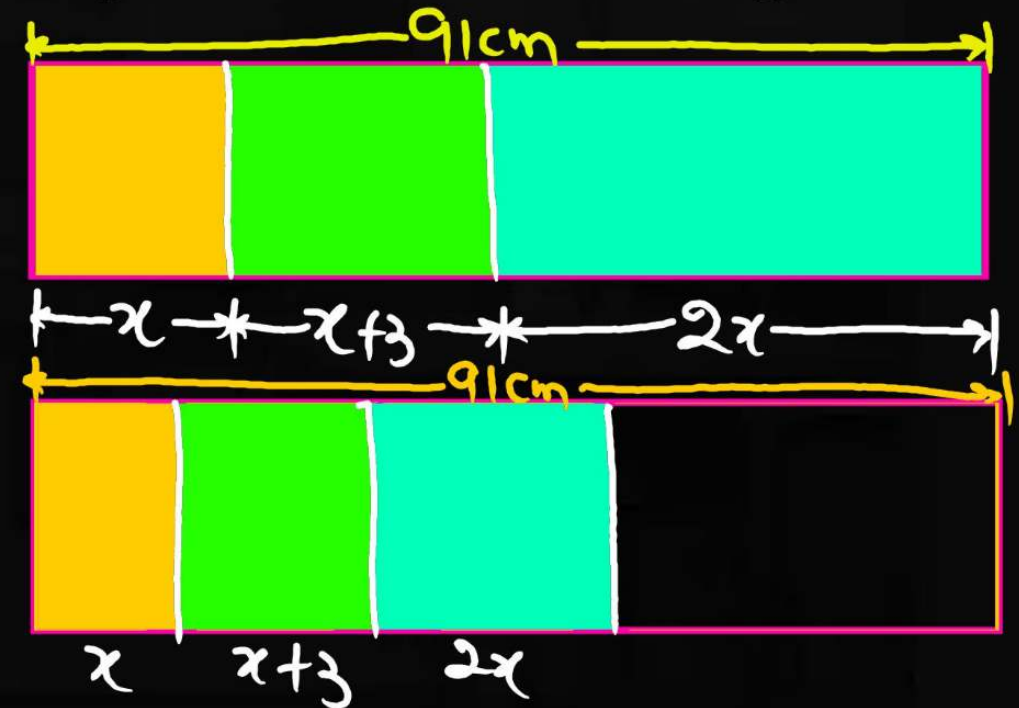
QUESTION

A man wants to cut three lengths from a single piece of board of length 91cm. The second length is to be 3cm longer than the shortest and the third length is to be twice as long as the shortest. What are the possible lengths of the shortest board if the third piece is to be at least 5cm longer than the second?

[Hint: If x is the length of the shortest board, then x , $(x + 3)$ and $2x$ are the lengths of the second and third piece, respectively.

Thus, $x + (x + 3) + 2x \leq 91$ and $2x \geq (x + 3) + 5$.

(Sol) let the shortest length = x cm
 $\Rightarrow$ second length = $(x + 3)$ cm
 & third length = $2x$ cm



Then, according to the question;

$$\begin{aligned} 2x &\geq (x+3) + 5 \\ \Rightarrow 2x &\geq x + 8 \\ \Rightarrow 2x - x &\geq 8 \\ \Rightarrow x &\geq 8 \quad \text{--- (1)} \end{aligned}$$

$$\begin{aligned} \text{Also, } x + (x+3) + 2x &\leq 91 \\ \Rightarrow 4x + 3 &\leq 91 \\ \Rightarrow 4x &\leq 88 \\ \Rightarrow x &\leq \frac{88}{4} \\ \Rightarrow x &\leq 22 \quad \text{--- (2)} \end{aligned}$$

From (1) & (2), we get

$$8 \leq x \leq 22$$

Therefore the possible lengths of the shortest piece is greater or equal to 8cm & less or equal to 22cm

Hence the result.

Thank
You